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Research proposal presentation

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Research question

How does instruction placement in long prompts affect the accuracy of Qwen3 models, and can thinking architecture reduce the “Lost in the Middle” effect?

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Importance

Large Language Models are used in research, education, and general decision support. Often using long prompts by providing context.

Prior research shows the **“Lost in the Middle”** phenomenon, where instructions placed in long prompts are partially ignored (Liu et al., 2024).

Understanding how instruction placement**,** model size, and model architecture affect response accuracy is critical for improving prompt engineering practices, selecting appropriate models for long-context tasks, etc.

This study compares **Thinking vs. Instruction-tuned models** to provide insights into when and why prompt structure matters.

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Literature review - variables

- Independent variables:
  - Prompt structure:
    - Provide context of a historical event at 700 - 1000 words at length
    - Create an instruction about the historical event
    - Insert the instruction in either start, middle or end
  - Model size - 8B, 30B or 235B parameter count
  - Model architecture - Thinking or Instruct
- Dependent variables
  - Accuracy - whether the answer contained the correct response

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Literature review - Controlling variables

- Temperature - 0
  - Randomness for token generation
- Seed - 42
  - Initial number for random token generation
- Top_p - 1
  - Dynamic cutoff for the most probable words
- Top_k - 1
  - Hard cutoff for most probable words (will always take the best one)
- Min_p - 0
  - Discards any token that is less than X% as probable as most probable token
- Thinking (exclusive to thinking models) - medium

Literature review - Controlling variables

- Repetition_penalty - 1
- Frequency_penalty - 0
- Presence_penalty - 0
  - All of them control how much words can repeat in its answers
  - Differ by mathematical calculation of penalty
- Logit_bias - null
  - Manual control of words that can or cannot appear in the answers
- Max_tokens - 2000
  - Hard cutoff for the answer

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Theoretical explanation

- Prompt structure
  - Expected a U shaped correlation with accuracy since models have primacy and recency bias
- Model size
  - The larger the model the higher the accuracy
- Model architecture
  - Thinking models will be more accurate as they synthesize their “thought” process

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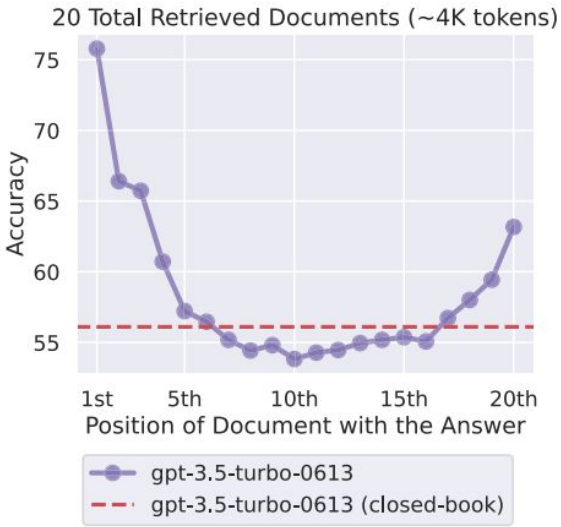
Empirical explanation - Prompt Structure

Figure 1

Lost in the middle phenomenon



Note. The figure illustrates the "lost in the middle" phenomenon, where the accuracy of the LLMs varies depending on the position of the relevant information within the prompt.

- LLMs are trained on data where most useful information comes first
  - Thus causing primacy bias
- LLMs predict next token based on its previous tokens
  - Thus putting more emphasis on most recent tokens causing recency bias
- LLMs are trained to process the text entirely and then to answer the prompt
  - Thus middle instructions are diminished by surrounding context

(Liu et al., 2024)

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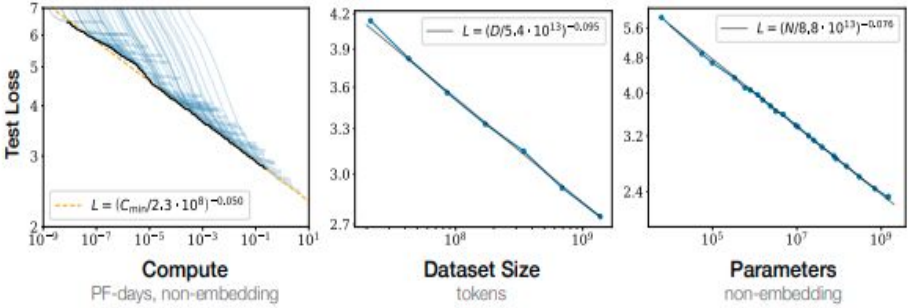
Empirical explanation - Model Size

Figure 2

Power of scaling law



Note. The figure illustrates as the compute, dataset and parameter size is increased the models performance increases

- The reason why scaling law works is not fully clear
- Hypothesized, that at large scale model learns to detect increasingly subtle patterns
- Compared models from 768 to 1.5B parameters
- The larger the model, the better the performance

(Kaplan et al., 2020)

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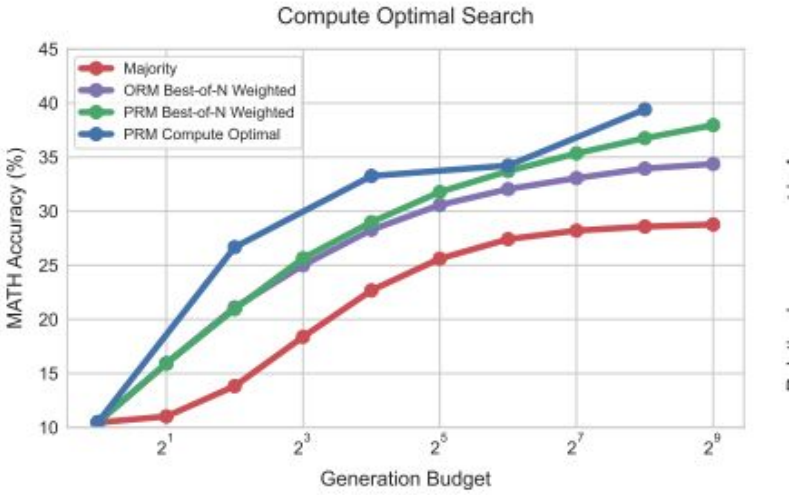
Empirical explanation - thinking

Figure 3

LLM reasoning improvements



Note. The figure illustrates as the Generation Budget ("Time") increases the accuracy increases as well

- Depicts that LLM reasoning improved math results by 4 times
- Found that a smaller model reasoning more often outperforms a 14x larger pre-trained model
- Forces model to revisit the problem and find mistakes or apply different strategies

(Snell et al., 2024)

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Participants

The research compares multiple Qwen3 VL large language models, including both Instruction-tuned and Thinking architectures.

Models vary by **parameter size** (8B, 30B, 235B) and **model version** (older and newer releases)

- Qwen3 VL 8B Thinking
- Qwen3 VL 8B Instruct
- Qwen3 VL 30B A3B Thinking
- Qwen3 VL 30B A3B Instruct
- Qwen3 VL 235B A22B Thinking
- Qwen3 VL 235B A22B Instruct

Materials and stimuli

The primary materials consist of text-based prompts describing historical events. Each prompt contains a 700–1000 word contextual passage followed by a task instruction.

Prompt structure will change based on the instruction placement:

- In the beginning
- The middle
- The end

Prompt content is initially generated using a separate AI model and then **manually edited** to ensure consistency, clarity, and relevance across all conditions.

Measurements

Model performance will be evaluated based on whether it contains the correct answer to the given instruction.

Accuracy is measured with:

1 - response satisfies the expected criteria

0 - response does not satisfy the expected criteria

Results are compared across the following factors:

- instruction placement (start vs. middle vs. end)
- model size
- model architecture (Thinking vs. Instruction-tuned)

Procedure

Generate at least **15 prompts** about different historical events (*distinct by topics*) $\longrightarrow$

For each event, a base prompt is created and modified into three prompt structures with instruction placement at the beginning, middle, or end. $\longrightarrow$

Each model receives 45 prompts $\longrightarrow$

Responses are analyzed to determine accuracy.

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References

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Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics, 12*, 157–173. https://doi.org/10.1162/tacl_a_00638

Snell, C., Lee, J., Xu, K., & Kumar, A. (2024). *Scaling LLM test-time compute optimally can be more effective than scaling model parameters*. arXiv. https://arxiv.org/abs/2408.03314